

INFORMATION ABOUT THE COVER

IAPMO and its industry partners have long promoted the benefits of industry developed codes and standards. With this issue, IAPMO and the Western Fire Chiefs Association (WFCA) introduces a new design for the Uniform Codes. The design ensures that Uniform Codes will have a distinctive and meaningful appearance. The new design contains various features that we hope will add to the value of these tried and tested codes.

The “rainbow” is called the Uniform Code Spectrum and represents that the Uniform Code family will cover the entire spectrum of construction codes, all maintained under a cohesive umbrella and all providing for cross-representation and correlation between industry professionals and interested participants. The Uniform Code Spectrum extends onto the spine of the book and distinctively allows these books to be immediately located among hundreds of other titles.

Each individual code has a distinctive color which will allow users through the years to find their own book among a shelf full of Uniform Codes. For example, the plumbing code is blue, the mechanical code is green and the fire code is red. This color is reflected both as the predominant background of the book and is also contained as one of the color bands in the spectrum.

The Uniform Code cover also contains an embedded graphic which evokes scenes from the code professional’s experience. This graphic is itself subdivided, with one portion being identical among all the Uniform Codes and another portion being unique to the particular code. Please join us in celebrating the new millennium by supporting the continued development and maintenance of the Uniform Codes.

FAMILY OF CONSENSUS CODES

At IAPMO’s 70th Annual Education and Business Conference, held in Washington, DC, the International Association of Plumbing and Mechanical Officials and the National Fire Protection Association executed an historic agreement. This agreement will provide for an integrated family of consensus codes to be published and maintained by IAPMO and NFPA for use in all sectors of the construction industry including, but not limited to plumbing, mechanical, fire, electrical and life safety. These codes will be developed and maintained to protect our world community from harmful diseases passed by unsanitary conditions.

This agreement signifies a commitment by IAPMO and NFPA to ensure the safe installation of systems to protect our nations citizens. In national and international arenas, both IAPMO and NFPA will work side-by-side to promote the most inclusive and globally respected codes for all mankind. This family of integrated consensus codes will be available with the printing of the 2003 editions.

UNIFORM PLUMBING CODE™

2000 EDITION



Adopted at the Seventieth Annual Conference
September, 1999

INTERNATIONAL ASSOCIATION OF PLUMBING AND
MECHANICAL OFFICIALS
A Nonprofit Association



REVISION MARKINGS

Code changes from the 1997 edition are marked in the margins as follows.

← An arrow denotes a deletion

| A vertical line denotes a change

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FOREWORD

The advantages of a uniform plumbing code adopted by various local jurisdictions have long been recognized. Disorder in the industry as a result of widely divergent plumbing practices and the use of many different, often conflicting, plumbing codes by local jurisdictions influenced the Western Plumbing Officials Association (now the International Association of Plumbing and Mechanical Officials [IAPMO]) to form a committee of plumbing inspectors, master and journeyman plumbers, sanitary and mechanical engineers, assisted by public utility companies, and the plumbing industry to create a basic plumbing document for general use. The product of this effort, the first edition of the Uniform Plumbing Code™ (UPC™) was officially adopted by IAPMO in 1945. The widespread use of this code over the past five decades by jurisdictions throughout the United States is testimony to its merit.

With the publication of the 1997 edition of the UPC, for the first time in history, a plumbing code has been created which is the result of a collaboration of industry-wide entities. The National Association of Plumbing-Heating-Cooling Contractors (NAPHCC) and the Mechanical Contractors Association of America (MCAA) joined forces with IAPMO to create this amalgamation of the most desirable aspects of the three most respected plumbing codes in existence. The 1994 Uniform Plumbing Code published by IAPMO, the 1993 ANSI A40 Safety Requirements for Plumbing™ published by the MCAA and NAPHCC Joint Task Force, and the National Standard Plumbing Code™ published by NAPHCC were scrupulously dissected and reassembled into this, the finest plumbing code document ever published.

With the publication of this 2000 edition of the UPC another significant milestone has been reached. The American Society of Sanitary Engineers (ASSE), United Association of Journeyman and Apprentices of the Plumbing and Pipe Fitting Industry of the United States (UA), and the Western Fire Chiefs Association (WFCA) have endorsed the use of the UPC as the national plumbing for use in this county and internationally.

IAPMO and the National Fire Protection Association (NFPA) announced at the September, 1999 IAPMO Conference they will work jointly to develop IAPMO codes and standards using the

NFPA/ANSI consensus model. The 2003 edition of the IAPMO UPC and UMC will be coordinated and harmonized with all NFPA Standards.

The memberships of the aforementioned associations are composed of representatives of all facets of the plumbing industry. They include plumbing contractors, installers, inspectors, building officials, engineers, architects, designers, manufacturers, wholesalers and consumers.

The UPC is designed to provide consumers with safe and sanitary plumbing systems, while, at the same time, allowing latitude for innovation and new technologies. The users of the UPC are continuously encouraged to update and to submit changes to improve the code. Amendments adopted by the IAPMO membership are incorporated into the code, which is published every three years. It is this process that keeps the code current with the latest technological advances in the industry.

The Uniform Plumbing Code is dedicated to all those who, in working to achieve “the ultimate plumbing code,” have unselfishly devoted their time, effort, and personal funds to create and maintain this, the finest plumbing code in existence.

The 2000 Uniform Plumbing Code is sponsored by ASSE, NAPHCC, MCAA, UA and WFCA. The presence of these logos, while reflecting support, does not imply any ownership of the copyright to the UPC which is held exclusively by IAPMO. The ASSE “service mark and logo” on the cover of this document indicates ASSE’s support of the voluntary, open, consensus process being used by IAPMO and NFPA to develop their codes and standards.

The addresses of endorsers and sponsors are as follows:

ASSE—28901 Clemens Road, Suite 100, Westlake, Ohio 44145, 440-835-3040.

NAPHCC—P.O. Box 6808, Falls Church, Virginia 22046, 800-533-7694.

NFPA—1 Batterymarch Park, Quincy, Massachusetts 02269, 617-770-3000.

MCAA—1385 Piccard Drive, Rockville, Maryland 20850, 301-869-5800.

UA — 901 Massachusetts Avenue NW, Washington D.C. 20001, 202-628-5823

WFCA—300 North Main Street, Suite 25, Fallbrook, California 92028, 760-723-6911.

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RELATED PUBLICATIONS AND SERVICES

IAPMO provides a variety of other products which are useful for inspectors, building officials, architects, engineers, manufacturers, contractors, plumbers, and apprentices.

IAPMO ORDER DESK
20001 WALNUT DRIVE SOUTH
WALNUT, CA 91789-2825
PHONE: 909-595-8449, EXT. 113, FAX: 909-594-3690
FOR PUBLICATIONS ORDERS: 800-85-IAPMO

Uniform Plumbing Code — Spanish Edition:

This UPC will assist the Spanish-speaking community in its utilization of the most widely adopted plumbing code in the world. The UPC was translated in response to numerous requests from jurisdictions throughout the US with large Spanish-speaking populations.

Uniform Plumbing Code on CD Rom:

With this CD, the Code is fully indexed and searchable, making it easy to get to the critical information you need without leafing through bulky books or dealing with torn pages.

Uniform Plumbing Code Interpretations Manual:

This increasingly popular manual is the result of ongoing work by IAPMO's Code Interpretations Committee. Updated annually, it contains hundreds of questions and answers. Many of these questions arise in the daily administration and enforcement of the code. Interpretation requests are accepted from active IAPMO members. Interpretation Request Forms are available in Official magazine or from IAPMO.

Uniform Plumbing Code Illustrated Training Manual:

The UPC Illustrated Training Manual is an excellent reference for anyone involved in the plumbing industry. It has an extensive definitions section and several hundred comprehensive technical diagrams and illustrations. It serves as a textbook, and it also is useful as a valuable tool for explaining the intent and use of the Code.

Uniform Plumbing Code Study Guide:

This book is the perfect complement to the UPC Illustrated Training Manual. Alone, it constitutes a complete chapter-by-chapter self-study course for learning the UPC. It has hundreds of questions, general practice exams, a section on plumbing math, numerous pipe sizing exercises, and a section on fitting identification. The questions and answers in the UPC Study Guide are kept current by IAPMO's Education Committee.

Dwelling Requirements of the Uniform Plumbing Code:

This book contains the UPC requirements which apply to one and two family dwellings only. Its illustrations and photographs make it especially useful for combination dwelling inspectors and the do-it-yourselfer.

Cumulative Analysis of Uniform Plumbing Code Changes:

An excellent reference for learning and understanding plumbing code changes adopted at the Annual Education and Business Conferences. It is also useful for parties preparing proposed code changes.

Uniform Mechanical Code

The Uniform Mechanical Code contains complete requirements for the installation and maintenance of heating, ventilating, cooling, and refrigeration systems.

Uniform Mechanical Code Study Guide:

Similar in design to the Uniform Plumbing Code Study Guide, the Uniform Mechanical Code Study Guide covers all aspects of the UMC in detail.

Handbook to the Uniform Mechanical Code

An indispensable tool for understanding the provisions of the Uniform Mechanical Code.

Uniform Solar Energy Code:

Provides a complete set of regulations for the use of solar energy in both plumbing and mechanical systems.

Uniform Swimming Pool, Spa, and Hot Tub Code:

Contains complete and current guidelines for piping systems serving these increasingly popular amenities.

IAPMO Installation Standards:

IAPMO standards committees have formulated installation standards for a wide variety of commonly used plumbing materials and systems. The IAPMO installation standards are included after the text of the Uniform Plumbing Code. Separate printed copies are also available.

Material and Property Standards:

Although IAPMO does not generally develop material and property standards, when it becomes apparent that necessary standards are nonexistent, IAPMO has exercised leadership by developing them. IAPMO's standards are always subject to modifications and improvements, and are withdrawn when acceptable nationally-recognized consensus standards are formulated.

Manufactured Home and Recreational Vehicle Standards:

Similar to the material and property standards, IAPMO has developed a number of standards for specialty products used in manufactured homes and recreational vehicles. These standards are available individually, or they may be purchased as a set.

ANSI Z 124 Standards:

These standards are written to specifically address a variety of plastic plumbing fixtures and components. They are available individually, or may be grouped for quantity discounts.

Directory of Listed Plumbing Products:

The IAPMO R&T Listing Program was primarily created to benefit the building and plumbing officials in the growing number of jurisdictions that adopt and use the UPC. For a product to be listed by IAPMO R&T, it must be found to be in conformance with an accepted standard or criteria and must not be in conflict with the UPC. IAPMO R&T administers a listing compliance program with unannounced inspections of manufacturing facilities, conducted to verify compliance with listing requirements.

This directory contains information on several thousand IAPMO R&T listed plumbing products and is updated monthly.

Directory of Listed Plumbing Products for Manufactured Housing and Recreational Vehicles:

This directory contains information on several hundred IAPMO R&T listed plumbing products used in manufactured housing and recreational vehicles and is updated bi-monthly.

Official Magazine:

IAPMO's bimonthly publication features informative articles on plumbing, local chapter proceedings and events, and plumbing industry updates. All IAPMO members receive a free subscription.

Drain Waste and Vent Calculator:

This calculator provides quick and simple access to the fixture unit and sizing tables of Chapter 7 of the UPC.

Water Sizing Calculator:

Sizing water systems becomes a much simpler task with the use of this handy calculator.

Natural Gas Pipe Sizing Calculator:

Designed for systems with a supply pressure of six to eight inches of water column, all pipe capacities are given in cubic feet per hour.

MEMBERSHIP IN IAPMO

IAPMO is an open association. Membership categories have annual dues from \$10.00 to \$300.00. Member benefits include discounted prices on most IAPMO publications, a subscription to Official magazine, and the opportunity to participate in updating and improving the most comprehensive plumbing code available.

Some membership categories include additional benefits. For additional information, contact IAPMO's Membership Department.

ORGANIZATION OF IAPMO

IAPMO is a nonprofit association governed by a Board of Directors, consisting of a President, Vice President, Immediate Past President, and nine Directors. The Board of Directors is elected by the active members at the Annual Education and Business Conference. The President and Vice President serve one year terms. The Directors serve for three years. A Secretary and Treasurer are appointed annually from the active membership by the Board of Directors.

The daily business of the association is conducted by IAPMO's staff, in coordination with IAPMO's standing and ad hoc committees.

UPC CODE CHANGE CYCLE

For the first time in IAPMO's long history, the 2003 edition of the Uniform Plumbing Code (UPC) will be developed utilizing a consensus process. The UPC will be harmonized with the Uniform Mechanical Code (UMC) and with the National Fire Protection Associations' National Electrical Code, Life Safety Code and Fire Code.

HOW TO CONTACT IAPMO

Mailing Address:

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Publications orders:
1-800-85-IAPMO

Office Hours:

Monday through Friday
8:00 am to 5:00 pm Pacific time
Closed weekends and holidays

Internet Information:

Web Site: <http://www.iapmo.org>
E-Mail Address: iapmo@iapmo.org

Sample Ordinance for Adoption of the 2000 Uniform Plumbing Code

Ordinance No. ____

An ordinance of the ____ (jurisdiction) adopting the 2000 edition of the "Uniform Plumbing Code", including IAPMO Installations Standards contained in Appendix I, regulating and controlling the design, construction, quality of materials, erection, installation, alteration, repair, location, relocation, replacement, addition to, use or maintenance of any plumbing system in the ____ (jurisdiction); providing for the issuance of permits and collection of fees therefore; repealing Ordinance No. ____ of the ____ (jurisdiction) and all other ordinances and parts of the ordinances in conflict therewith.

The ____ (governing body) of the ____ (jurisdiction) does ordain as follows:

Section 1. That certain documents, three (3) copies of which are on file in the office of the ____ (jurisdiction's keeper of records) and the ____ (jurisdiction), being marked and designated as "Uniform Plumbing Code", (including Appendix chapters ____ (fill in the applicable Appendix chapters) be and are hereby adopted as the code of the ____ (jurisdiction) for regulating the design, construction, quality of materials, erection, installation, alteration, repair, location, relocation, replacement, addition to, use or maintenance of plumbing systems in the ____ (jurisdiction) providing for the issuance of permits and collection of fees therefore; and each and all of the regulations, provisions, conditions and terms of such "Uniform Plumbing Code", 2000 edition, and Appendix I published by the International Association of Plumbing and Mechanical Officials, on file in the office of the ____ (jurisdiction) are hereby referred to, adopted and made a part hereof as if fully set out in this ordinance.

Section 2. (Incorporate penalties for violations. See Section 102.3.2.)

Section 3. That Ordinance No. ____ of ____ (jurisdiction) entitled (fill in here the complete title of the present plumbing ordinance or ordinances in effect at the present time so that they will be repealed by definite mention) and all other ordinances or parts of ordinances in conflict herewith are hereby repealed.

Section 4. That if any section, subsection, sentence, clause or phrase of this ordinance is, for any reason, held to be unconstitutional, such decision shall not affect the validity of the remaining portions of this ordinance. The ____ (governing body) hereby declares that it would have passed this ordinance, each section, subsection, clause or phrase thereof, irrespective of the fact that any one or more sections, subsections, sentences, clauses and phrases be declared unconstitutional.

Section 5. That the ____ (jurisdiction's keeper of records) is hereby ordered and directed to cause this ordinance to be published. (An additional provision may be required to direct the number of times the ordinance is to be published and to specify that it is to be in a newspaper in general circulation. Posting may also be required.)

Section 6. That this ordinance and the rules, regulations, provisions, requirements, orders and matters established and adopted hereby shall take effect and be in full force and effect ____ (time period) from and after the date of its final passage and adoption.